

Comparison of Rectifier Circuits

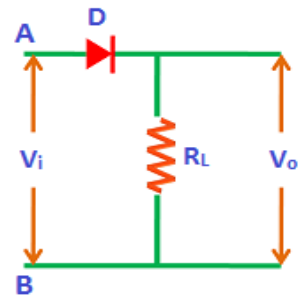
Parameter	HWR	FWR (Center-Tap Rectifier)	Bridge Rectifier
Circuit Diagram			
Number of Diodes Used	One	Two	Four
Center Tap Transformer	Not Required	Very Much Required	Not Required
Transformer Core Saturation	Possible	Not Possible	Not Possible
Operation During Positive Half Cycle (0 - π)			
Operation During Negative Half Cycle (π - 2π)			
Waveform			
Average Load Volatge (V_{Ldc})	$\frac{V_m}{\pi}$	$\frac{2V_m}{\pi}$	$\frac{2V_m}{\pi}$
Average Load Current (I_{Ldc})	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
RMS Load Volatge (V_{Lrms})	$\frac{V_m}{2}$	$\frac{V_m}{\sqrt{2}}$	$\frac{V_m}{\sqrt{2}}$
RMS Load Current (I_{Lrms})	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
DC Load Power (P_{dc})	$\frac{I_m^2}{\pi^2} R_L$	$\frac{4I_m^2 R_L}{\pi^2}$	$\frac{4I_m^2 R_L}{\pi^2}$
PIV (Peak Inverse Voltage)	V_m	$2V_m$	V_m
Peak Load Current (I_m)	$I_m = \frac{V_m}{(R_s + R_F + R_L)}$	$I_m = \frac{V_m}{(R_s + R_F + R_L)}$	$I_m = \frac{V_m}{(R_s + 2R_F + R_L)}$
Maximum Rectification Efficiency (η)	40%	81.20%	81.20%
TUF (Transformer Utilation Factor)	28.70%	69.30%	81.20%
Ripple Factor $RF = [FF^2 - 1]^{1/2}$	121%	48%	48%
Ripple Frequency	50 Hz	100 Hz	100 Hz

Clipper Circuits

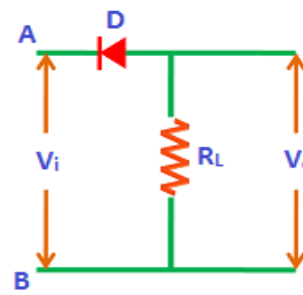
Types →

Unbiased Clippers

Seies Negative Clipper

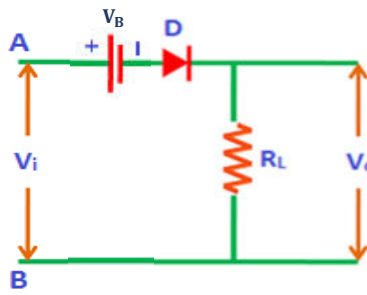


Series Positive Clipper

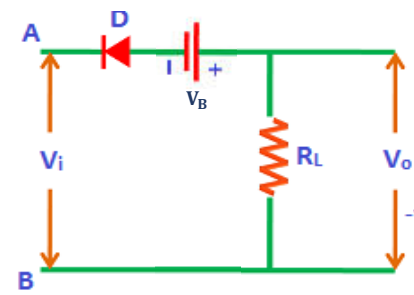


Biased Clippers

Seies Negative Clipper

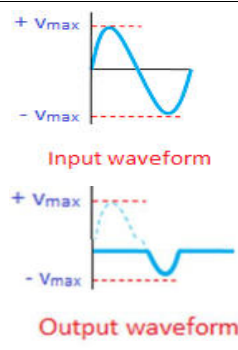
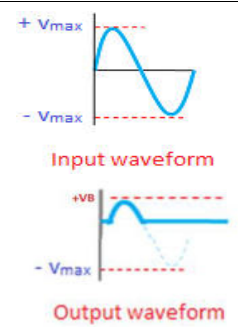
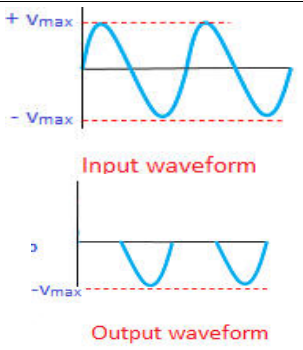
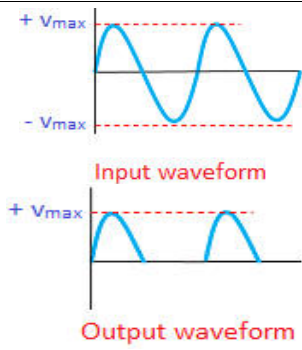


Series Positive Clipper



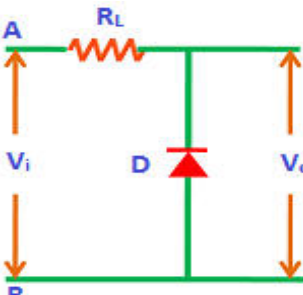
Equivalent
Circuit

Input Output
Waveform

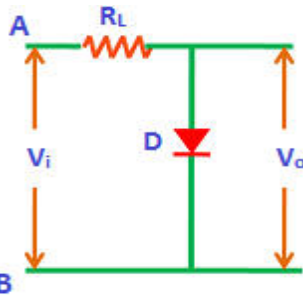


Types →

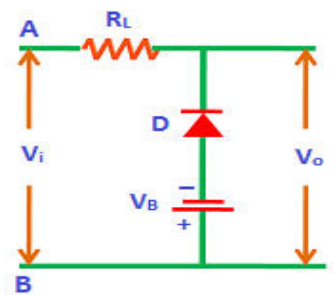
Parallel Negative Clipper



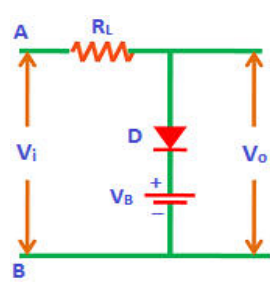
Parallel Positive Clipper



Parallel Negative Clipper

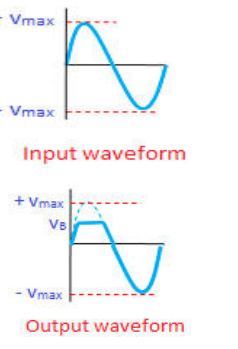
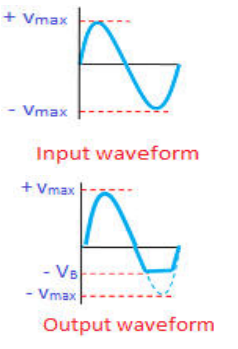
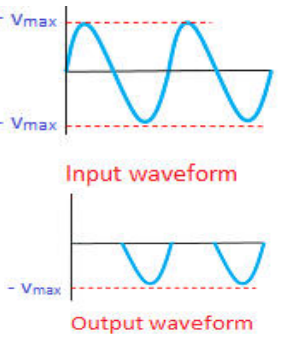
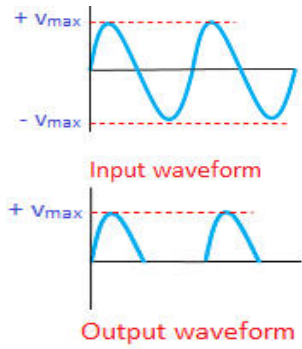


Parallel Positive Clipper

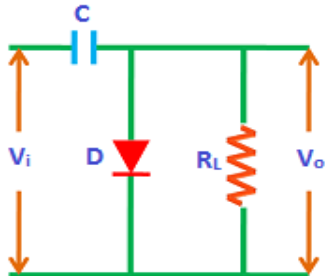
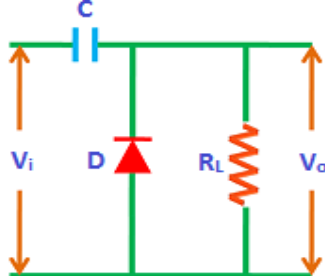
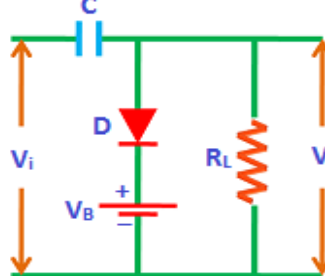
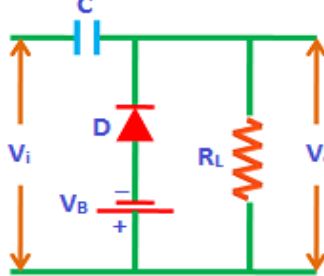
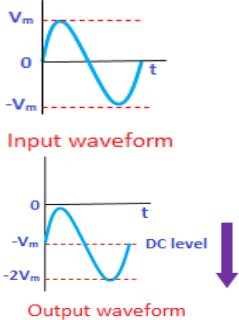
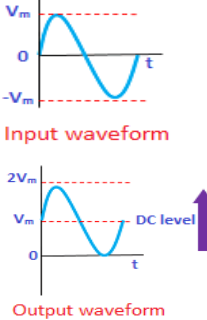
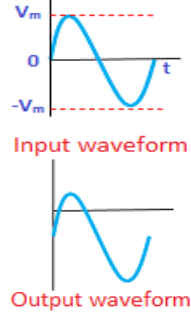
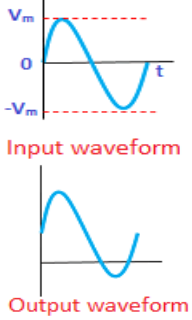


Equivalent
Circuit

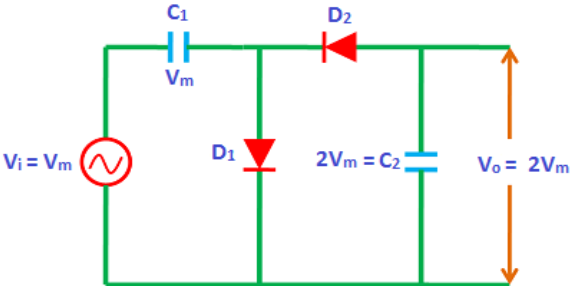
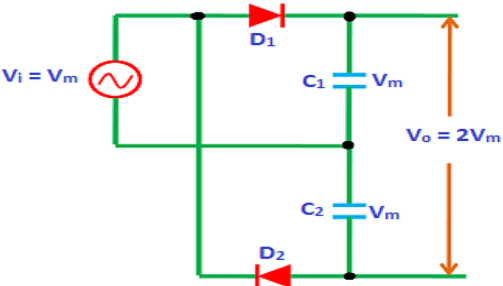
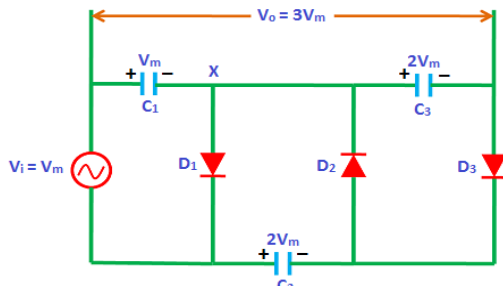
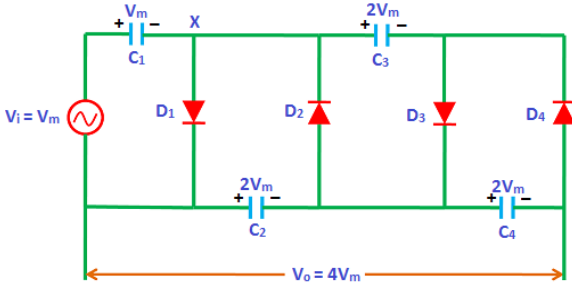
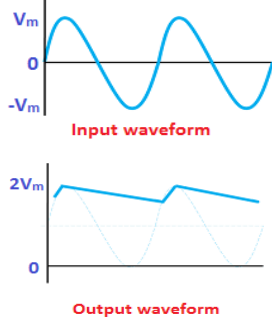
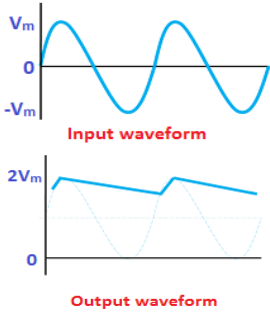
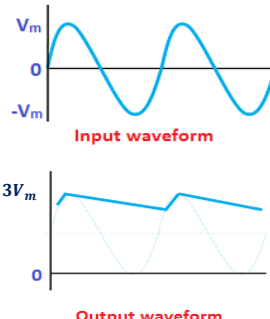
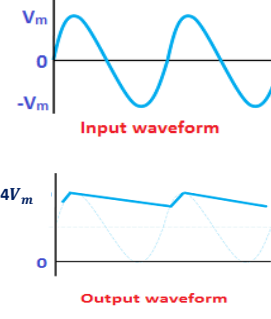
Input Output
Waveform



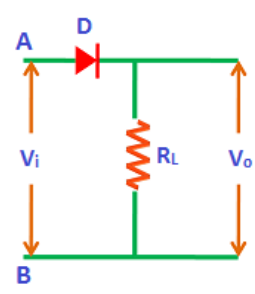
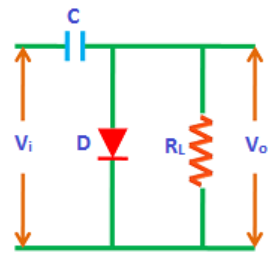
Clamper Circuits

Types →	Unbiased Clamper		Biased Clamper	
	Negative Clamper	Positive Clamper	Negative Clamper	Positive Clamper
Equivalent Circuit				
Input Output Waveform				

Volatge Multipliers Circuits

Types →	Half Wave Voltage Doubler	Full Wave Voltage Doubler	Full Wave Voltage Tripler	Full Wave Voltage Quadrupler
				
Input Output Waveform				

Comparison between Clipper and Clamper

Parameter	Clipper	Clamper
Function	To remove a part of input waveform.	To add a DC shift to the input waveform.
Component Used	Diode(D), Resistors (R)	Diode(D), Resistors (R), Capacitor (C)
Energy storage component	Not required	Requires (Capacitor (C) is used as energy storage element)
Output Voltage	Less than the input voltage.	Multiples of input voltage.
DC Level	Remains same	DC level get shifted
Frequency of Input	Not important as capacitor is not used.	The value of C needs to be chosen on the basis of input frequency.
Application	In transmitters, receivers, amplitude selector, noise limiter etc.	Voltage Multipliers, Radar system etc.
Configuration	 Series Negative Clipper	 Negative Clamper